

Nonlinearity-Induced Spur Analysis in Fractional- N Synthesizers with $\Delta\Sigma$ Quantization Cancellation

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Abstract—A fractional- N frequency synthesizer with low total jitter [e.g., $<50\text{ fs}_{\text{rms}}$, accounting for both phase noise (PN) and spurs] is essential for enabling the emerging 5G/6G and other high-speed wireless communication standards (e.g., WiFi-6/7). While fractional- N phase-locked loops (PLLs) and injection-locking techniques with delta-sigma ($\Delta\Sigma$) quantization cancellation using a digital-to-time converter (DTC) (and, more recently, DACs) have demonstrated low-jitter performance and are well-understood in terms of PN, their spur mechanisms still lack a comprehensive quantitative analysis. In this paper, we present a unified theoretical framework for spur analysis, based on the time-domain characteristics of spurs, addressing both instantaneous phase modulation and frequency modulation mechanisms. This approach serves as a thorough guide for choosing a low-jitter fractional- N architecture, considering the integral nonlinearity (INL) shaping of DTCs (or DACs) under the control of either a 1st- or 2nd-order $\Delta\Sigma$ modulator (DSM). The framework also extends to reference spurs in both charge-pump PLLs (CP-PLLs) and injection-locked synthesizers. The analytical results of spurs are numerically verified through time-domain behavioral simulations and further validated by experimental results from the literature, thereby demonstrating their effectiveness.

Index Terms—Spur, digital-to-time converter (DTC), all-digital phase-locked loop (ADPLL), fractional- N , millimeter-wave (mmWave), delta-sigma modulator (DSM), integral nonlinearity (INL).

I. INTRODUCTION

THE emerging 5G/6G [1] and WiFi-6/7 [2] wireless communication standards require support for complex modulation schemes (e.g., 4096-QAM) with large bandwidth (e.g., 800 MHz) to meet the demand of ever-increasing data rates. This necessitates a fractional- N frequency synthesizer featuring low phase noise (PN), low jitter, and minimal spurs [3]–[5]. To achieve low jitter, two general techniques have been employed [6]: 1) a PLL with a high gain of its phase-detector (PD), and 2) injection locking (IL) [7] or charge-sharing locking (CSL) [8], [9]. A high PD gain can be achieved by sampling a *sharp* oscillator's edge in voltage-domain [10]–[13] or charge-domain [14], [15], sampling a reference's (or divider's) edge [16], [17] conditioned by a *sharp* RC filter [18], [19], or resorting to a bang-bang operation (i.e., 1-bit

TABLE I
EVM REQUIREMENTS FOR 5G/6G [1] AND WiFi-6/7 [2].

	5G/6G (FR2)	WiFi-6	WiFi-7
QAM	256	1024	4096
Required EVM	3.5%	1.77% (−35dB)	1.25% (−38dB)
EVM_{LO}	1.4%	0.71%	0.5%
EVM_{TX}	3.2%	1.62%	1.15%

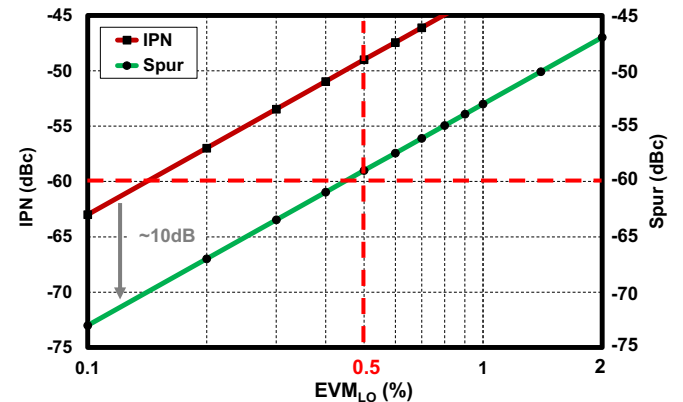


Fig. 1. Spur requirements to prevent degradation of the LO's integrated random phase noise (IPN) in the next generation of communications.

mid-rise quantizer) [20]–[25]. Nearly all of these techniques are *intrinsically* limited to an integer- N operation, relying on the precise edge alignment between the oscillator (or divider) and reference.

By using a digital delta-sigma ($\Delta\Sigma$) modulator (DSM)-controlled divider, divider-based PLLs enable the fractional- N operation [26], but they also introduce fractional- N spurs due to the significant quantization error [27]–[30]. To suppress such noise, “ $\Delta\Sigma$ quantization cancellation” techniques are employed to appropriately delay the reference edges (or divider edges [31]) by a digital-to-time converter (DTC) [18]–[20], [32], [33] to align with the divider edges (or reference edges). The DTC also enables the fractional- N operation in dividerless PLLs (such as counter-based [34]–[36] or sub-sampling [37], [38]) and injection-locked synthesizers [39] by aligning the reference edges directly with the oscillator edges. Besides, the phase error can be minimized in the voltage domain using a digital-to-voltage converter (e.g. DAC) in the reference path [40]–[46]. Given the above, the integral nonlinearity (INL) of DTC (or DAC) has become the primary contributor to the fractional- N spurs. To mitigate the adverse effects of these

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spurs on total jitter, the spur level must meet

$$\text{Spur}_{\text{dBc}} \leq \text{IPN}_{\text{dBc}} - 10 \text{ dB} \quad (1)$$

where IPN_{dBc} represents the integrated random phase noise. IPN_{dBc} can severely affect the error-vector magnitude (EVM), a key specification for a transmitter supporting complex modulation schemes. For the next-generation communications targeting 4096-QAM (see Table I), the EVM [2] contributed by the local oscillator (LO) (i.e., EVM_{LO}^1), is expected to be as low as 0.5%, accounting for around 16% of the total EVM's budget (i.e., $\text{EVM}_{\text{LO}}^2/\text{EVM}^2$). Consequently, the required spur level should be lower than -60 dBc , as shown in Fig. 1. Recently, a few promising techniques have emerged for suppressing the INL-induced spurs: 1) further manipulation of the probability density function (PDF) [29] of DTC's (or DAC's) control codes [32], [33], 2) DTC's INL compensation [45]–[48], and 3) high-linearity DTC techniques [49].

Nonetheless, quantitatively analyzing the INL-induced fractional- N spurs remains challenging and often requires time-consuming behavioral simulations followed by FFT analysis using numerical methods. Some initial estimation work for these spurs was presented in [50], [51], but it assumed a sinusoidal shape for the INL, which is convenient but typically insufficient. Moreover, this approach cannot account for INL-induced spurs when the DTC (or DAC) is controlled by a 2nd-order DSM [e.g., multi-stage noise shaping (MASH) 1-1], which introduces an additional code range [18], [31], [39], at the cost of doubling the DTC coverage range and generally worsening the INL. Clearly, a quantitative guide on the efficacy of 2nd-order DSM-based DTC controllers in fractional- N synthesizers is still lacking. In addition, equally important is to also accurately predict the reference spurs, as they not only can simply ruin the jitter [8] but may also mix with adjacent interferers, thereby worsening the receiver's SNR [52]. Specifically, the reference spurs in CP-PLLs have a nature of frequency modulation (FM), thus differing from the spurs' instantaneous phase modulation (PM) mechanism in DTC-based fractional- N and injection-locking synthesizers [8], [53], [54].

In summary, we recognize the need for a unified spur analysis method that can distinguish between the PM-induced and FM-induced spurs, calculate the level of fractional- N and reference spurs, and evaluate the DTC (or DAC)-based fractional- N spurs under different orders of DSM. In this paper, we discuss a quantitative spur analysis method based on the time-domain behavior, unifying the analysis of PM- and FM-induced spurs. In addition, we propose the new DTC INL-shaping modeling techniques that support spur analysis with both 1st- and 2nd-order DSM controllers. The rest of this paper is organized as follows: Section II presents the basic operating principles of the DTC-based fractional- N synthesizers with a 1st/2nd-order DSM. A general spur analysis technique in time-domain is discussed in Section III. Section IV introduces a quantitative analysis technique for fractional- N spurs using

¹ $\text{EVM}^2 = \text{EVM}_{\text{LO}}^2 + \text{EVM}_{\text{TX}}^2$, where EVM_{TX} represents the EVM contributed by the transmitter. If the spurs in the LO can be neglected, $\text{EVM}_{\text{LO}} = \sqrt{2} \cdot 10^{\text{IPN}_{\text{dBc}}/10}$.

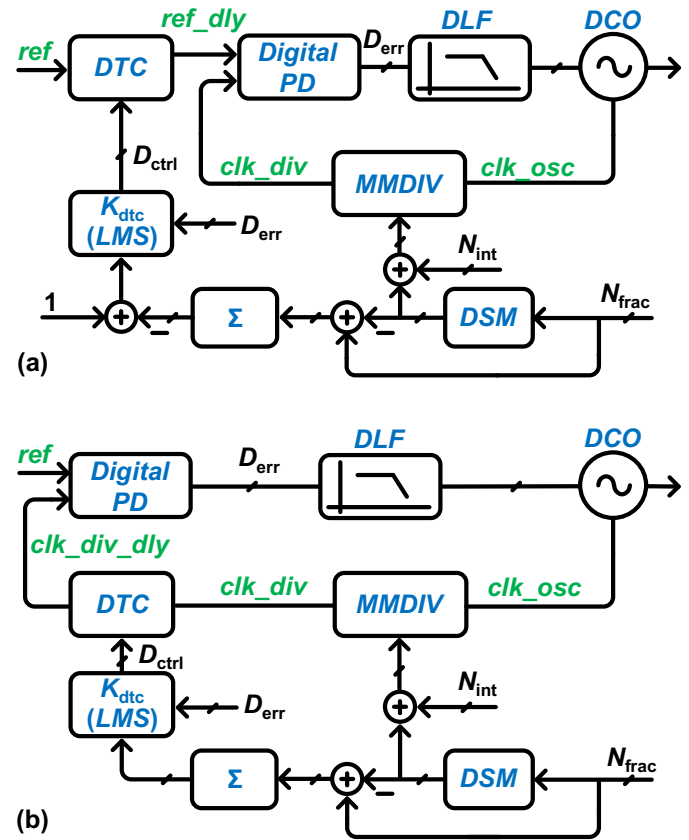


Fig. 2. Architecture of fractional- N digital PLLs utilizing the $\Delta\Sigma$ quantization cancellation technique, with the DTC placed in either (a) the reference path or (b) the divider path.

INL-shaping modeling. Reference spurs in both CP-PLLs and IL are discussed in Section V.

II. FRACTIONAL- N OPERATION WITH $\Delta\Sigma$ QUANTIZATION CANCELLATION TECHNIQUE

A. Architectures of Fractional- N Synthesizers with DTC

Fig. 2(a) illustrates a conceptual divider-based fractional- N digital PLL with a DTC in the reference path. The feedforward path consists of a digital phase detector (PD), a digital loop filter (DLF), and a digitally controlled oscillator (DCO). The feedback path comprises a multi-modulus divider (MMDIV) with a $\Delta\Sigma$ modulator (DSM) and other digital logic to enable the fractional- N operation. The DTC is employed to delay the reference signal ref to align it with the divided clock clk_div , thereby canceling the expected periodic fractional- N detecting error. Alternatively, the DTC can be placed in the feedback path to delay clk_div and align it with ref , as shown in Fig. 2(b). The DTC gain (K_{dte}) is typically calibrated using a least-mean-square (LMS) algorithm to minimize the fractional- N spurs. The DSM can be configured in either a 1st- or 2nd-order configuration (e.g., MASH 1-1 structure²), taking into account the INL and coverage range of the DTC.

²The integer output range of a 1st-order DSM is [0,1], whereas MASH 1-1 and MASH 1-1-1 extend the range to $[-1,2]$ and $[-3,4]$, respectively.

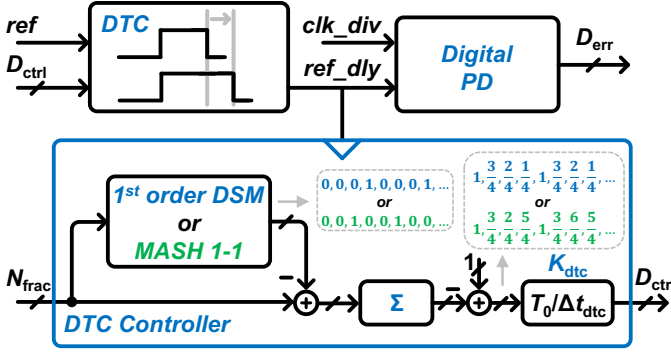


Fig. 3. Details of DTC controller with a 1st- or 2nd-order (e.g., MASH 1-1) DSM. Numerical example shown with $N_{\text{frac}} = 1/4$.

B. Fractional- N Operation with 1st-Order $\Delta\Sigma$ Modulator

To better explain the fractional- N operation in Fig. 2(a), the details of the DTC controller are disclosed in Fig. 3. We choose the rising edge of ref (of period T_{ref}) to update the DTC controller code-words, while its falling edge is controllably delayed by the DTC to align it with the falling edge of clk_div . As shown in Fig. 4(a), under a 1st-order DSM, the corresponding required delay can be expressed as

$$\Delta t_{\text{d,ref}}[n] = [-(nN_{\text{frac}} - \lfloor nN_{\text{frac}} \rfloor) + 1] \cdot T_0 \quad (2)$$

where n represents the corresponding discrete-time index, $T_0 = T_{\text{ref}}/N$ is the oscillator period, and N , comprising its integer part N_{int} and fractional part N_{frac} , is the PLL's frequency multiplication ratio. The offset “+1” (i.e., $+T_0$) is introduced to convert the required negative delay (i.e., advancing) into a positive one. The generated DTC code³ is $D_{\text{ctrl}}[n] = \Delta t_{\text{d,ref}}[n]/\Delta t_{\text{dtc}} = [-(nN_{\text{frac}} - \lfloor nN_{\text{frac}} \rfloor) + 1] \cdot K_{\text{dtc}}$, where $K_{\text{dtc}} (= T_0/\Delta t_{\text{dtc}})$ is the normalized DTC gain and Δt_{dtc} is the DTC's step-size resolution. If DTC is added in the divider path [31] for the purpose of delaying the falling edges of clk_div to align with the falling edges of ref , the required delay is given by $\Delta t_{\text{d,div}}[n] = (nN_{\text{frac}} - \lfloor nN_{\text{frac}} \rfloor) \cdot T_0$, as shown in Fig. 4(c).

C. Fractional- N Operation with 2nd-Order $\Delta\Sigma$ Modulator

To be able to further reduce the fractional- N spur level, the 1st-order DSM in the DTC controller can be amended by a 2nd-order DSM (e.g., MASH 1-1). In the time-domain, this approach ‘randomly’ adds the fixed T_0 delay into (2), as

$$\Delta t_{\text{d2,ref}}[n] = \Delta t_{\text{d,ref}}[n] + D_{\text{MASH}}[n] \cdot T_0 \quad (3)$$

where $D_{\text{MASH}}[n] = 1$ or 0 with a certain probability, shown in Fig. 3 and Fig. 4(b). However, this also extends the required DTC range from T_0 to $2T_0$, which will inevitably worsen the DTC INL⁴. Thus, there is a trade-off between using the 1st-

³The DTC controller in [34], [35] is a simplified version of that shown in Fig. 3, supporting only the 1st-order DSM fractional- N operation.

⁴If a 3rd-order DSM (e.g., MASH 1-1-1) were to be employed in Fig. 3 (with the offset “+1 (T_0)” replaced by “+2 (T_0)” just to avoid the negative DTC codes), the required DTC range would further increase to $4T_0$, which will significantly worsen the DTC INL and increase the design complexity. As a result, there are very few DTC-based $\Delta\Sigma$ quantization cancellation techniques reported that utilize a 3rd-order DSM.

or 2nd-order DSM quantization cancellation while considering the reduction of fractional- N spurs.

III. TIME-DOMAIN BEHAVIOR OF SPURS

A. Deterministic Jitter – General Spur Mechanism

To quantitatively study the spur mechanism, we begin by examining their behavior in time-domain, manifested as deterministic jitter. For a frequency synthesizer operating at frequency ω_0 , its output phase can be expressed as

$$\phi_{\text{out}}(t) = \omega_0 \cdot [t + \Delta t_{\text{RJ}}(t) + \Delta t_{\text{DJ}}(t)] \quad (4)$$

where $\Delta t_{\text{RJ}}(t)$ and $\Delta t_{\text{DJ}}(t)$ represent the instantaneous random jitter and deterministic jitter, respectively. For spurs at $\Delta\omega$ and its harmonics, the corresponding $\Delta t_{\text{DJ}}(t)$ can be expressed as

$$\Delta t_{\text{DJ}}(t) = \sum_{m=1}^M t_m \cos(m\Delta\omega t + \theta_m) \quad (5)$$

where t_m and θ_m represent the magnitude and phase of the m th harmonic, respectively. To calculate the spurious level, we substitute (5) into the fundamental component of the output waveform⁵:

$$\begin{aligned} v_{\text{out}}(t) &\approx V_{\text{H1}} \cos(\omega_0 t + \omega_0 \Delta t_{\text{DJ}}(t)) \approx V_{\text{H1}} \cos \omega_0 t \\ &+ \sum_{m=1}^M \frac{V_{\text{H1}} \omega_0 t_m}{2} \cos\left((\omega_0 + m\Delta\omega)t + \frac{\pi}{2} + \theta_m\right) \\ &+ \sum_{m=1}^M \frac{V_{\text{H1}} \omega_0 t_m}{2} \cos\left((\omega_0 - m\Delta\omega)t + \frac{\pi}{2} - \theta_m\right) \end{aligned} \quad (6)$$

where V_{H1} is the amplitude of the fundamental. Thus, the spur power level (unit: rad^2) at offset $m\Delta\omega$ is

$$S_m = \frac{(V_{\text{H1}} \omega_0 t_m / 2)^2 / 2}{V_{\text{H1}}^2 / 2} = |\pi \cdot t_m / T_0|^2 \quad (7)$$

which depends on the ratio between the m th harmonic magnitude of deterministic jitter Δt_{DJ} and the output period T_0 ($= 2\pi/\omega_0$). Typically, there are two types of periodic modulation mechanisms causing Δt_{DJ} and the corresponding spurs: 1) instantaneous phase modulation (PM) and 2) frequency modulation (FM), which should be distinguished. Interestingly, similar categories are found in the flicker phase noise [55] and phase correction mechanisms [6].

B. Spurs Due to Instantaneous Phase Modulation

The PM-induced deterministic jitter, $\Delta t_{\text{DJ,PM}}(t)$, can be written as

$$\Delta t_{\text{DJ,PM}}(t) = \frac{T_0}{2\pi} \Delta \phi_{\text{PM}}(t) \quad (8)$$

where $\phi_{\text{PM}}(t)$ is the periodically modulated phase. Thus, the PM-induced spur $S_{m,\text{PM}}$ can be directly calculated using (7) by conducting the analysis in the time domain. Typical PM-induced spurs include fractional- N spurs due to the DTC's nonlinearity or reference spurs in injection locking.

⁵ Δt_{RJ} can be neglected in the spur analysis.

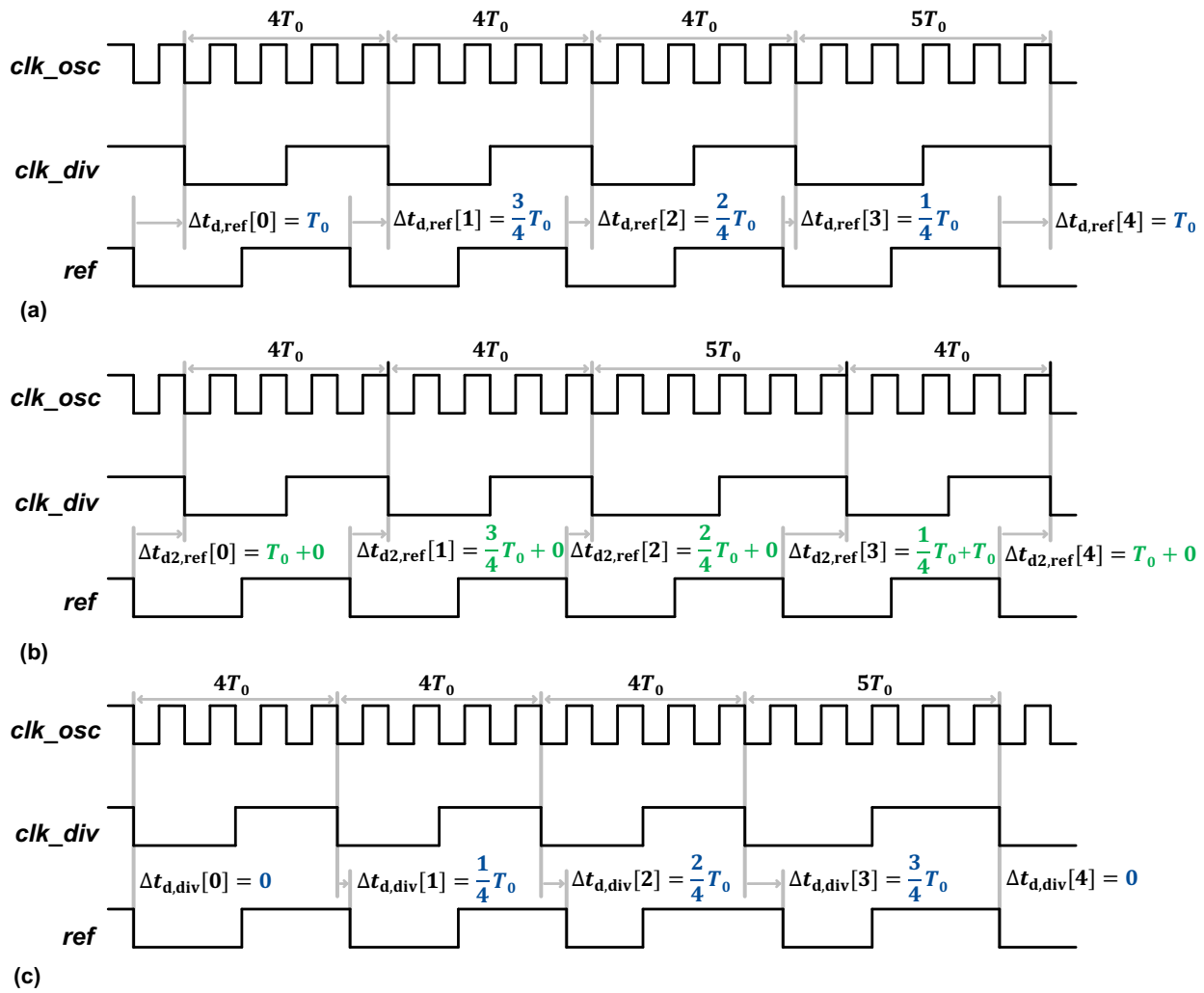


Fig. 4. Timing diagram of fractional- N operation with DTC in different configurations: (a) 1st-order DSM with DTC in the reference path, (b) 2nd-order DSM with DTC in the reference path, and (c) 1st-order DSM with DTC in the divider path. Examples with $T_{\text{ref}} = NT_0$, $N_{\text{int}} = 4$, and $N_{\text{frac}} = 1/4$.

C. Spurs Due to Frequency Modulation

Conversely, FM-induced spurs typically include reference spurs in CP-PLLs [52] and spurs caused by supply ripples acting on the oscillator's varactors and nonlinear parasitic capacitance of $-G_m$ devices [44]. As such, we can assume the periodically (i.e., at $\Delta\omega$) modulated frequency as

$$\Delta\omega_{\text{FM}}(t) = \sum_{m=1}^M \omega_m \cos(m\Delta\omega t + \theta_{\omega m}) \quad (9)$$

where ω_m and $\theta_{\omega m}$ are the magnitude and phase of the m th harmonic, respectively. The deterministic jitter can then be derived as

$$\begin{aligned} \Delta t_{\text{DJ,FM}}(t) &= \frac{T_0}{2\pi} \int_{-\infty}^t \Delta\omega_{\text{FM}}(\tau) d\tau \\ &= \sum_{m=1}^M \frac{T_0}{2\pi} \frac{\omega_m}{m\Delta\omega} \cos(m\Delta\omega t + \theta_{\omega m} - \frac{\pi}{2}) \end{aligned} \quad (10)$$

and the corresponding spur, per (7), is

$$S_{m,\text{FM}} = \left| \pi \cdot \frac{T_0}{2\pi} \frac{\omega_m}{m\Delta\omega} / T_0 \right|^2 = \left| \frac{1}{2} \frac{\omega_m}{m\Delta\omega} \right|^2. \quad (11)$$

Interestingly, the frequency-modulated spur is independent of the frequency synthesizer's period, T_0 . Furthermore, when $m = 1$, it degenerates into formula (1) in [44].

IV. QUANTITATIVE ANALYSIS OF FRACTIONAL- N SPURS

In fractional- N synthesis with a $\Delta\Sigma$ quantization cancellation, both the DTC's (or DAC's) INL and its finite resolution can introduce fractional- N spurs. However, the spurs resulting from limited resolution typically occur at much higher frequencies (e.g., $T_0/\Delta t_{\text{dtc}}$ times higher⁶) than those caused by the INL and are effectively filtered by the loop [56]. Additionally, to mitigate the spurs caused by the DTC's limited resolution, the DTC control codes D_{ctrl} can be 'rounded off' using a local 1st-order DSM (not shown in Fig 2), rather than simply disregarding (i.e., truncating) the fractional bits of D_{ctrl} [35].

⁶When N_{frac} is close to 0, the period of the INL-induced perturbation is $1/N_{\text{frac}} \cdot T_{\text{ref}}$, while the period of the quantization-induced perturbation is $\Delta t_{\text{dtc}}/(N_{\text{frac}} \cdot T_0) \cdot T_{\text{ref}}$.

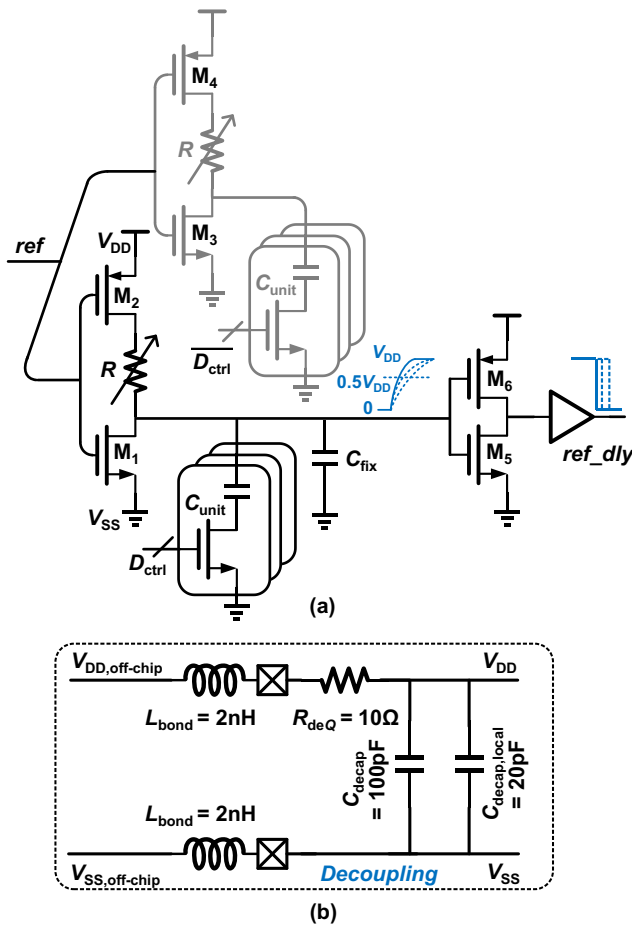


Fig. 5. (a) Schematic of RC-based DTC and (b) its on-chip supply decoupling scheme.

Consequently, we focus solely on the spurs induced by the DTC's (or DAC's) INL, which are located at

$$f_{frac} = \begin{cases} N_{frac} f_{ref}, & \text{if } N_{frac} < 0.5 \\ (1 - N_{frac}) f_{ref}, & \text{if } N_{frac} \geq 0.5 \end{cases} \quad (12)$$

and at higher harmonics (i.e., $m \cdot f_{frac}$). To quantitatively analyze these spur levels, it is essential to understand the INL modeling and how it is shaped under the control of a 1st- or 2nd-order DSM. In this section, we introduce an INL modeling technique for a popular RC-based DTC and conduct a comparative study of the induced fractional- N spurs using a 1st- and 2nd-order DSM controller. The proposed analysis methods can also be extended to other fractional- N synthesizers employing $\Delta\Sigma$ quantization cancellation.

A. Modeling INL of RC-Based Digital-to-Time Converter

A conventional RC-based DTC is depicted in Fig. 5(a). It consists of an input RC-delay stage, a dummy RC-delay stage [57], and a comparator. The input RC-delay stage delays the falling edges of ref via the RC charging, which includes $M_{1,2}$, a digitally controlled poly resistor R (mitigating the process corner's variation), a fixed capacitor C_{fix} (which can optimize the static INL to some extent by reducing

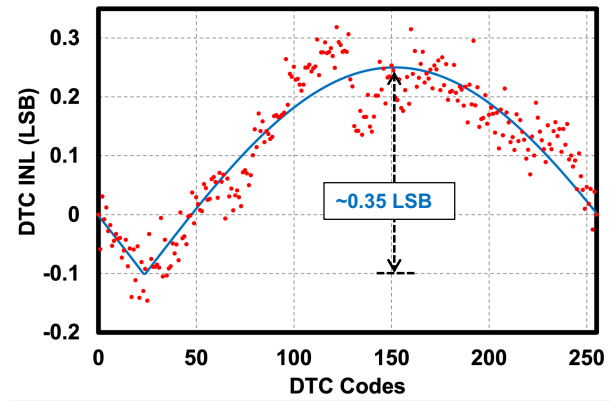


Fig. 6. Post-layout simulation of DTC's INL with consideration of L_{bond} , along with its fitting curve based on a "half-sine" shape ($\Delta t_{dte} \approx 500$ fs).

the code/slope-dependency of propagation delay with a flat slope) [18], and a K -bit C_{unit} -bank (controlled by D_{ctrl}). By adjusting the threshold voltage of the comparator (i.e., $M_{5,6}$) to $0.5V_{DD}$, the controlled delay of the DTC can be established as $0.69 \cdot R \cdot (D_{ctrl} \cdot C_{unit} + C_{fix})$. It is intrinsically proportional to the DTC control code D_{ctrl} , assuming $V_{DD} - V_{SS}$ remains relatively constant during the RC charging.

To stabilize the supply of $V_{DD} - V_{SS}$ and to reduce the dynamic INL of the DTC, the dummy RC-delay stage (controlled by the 1's complement of D_{ctrl} , i.e., $\overline{D_{ctrl}}$) is added, ensuring that V_{DD} charges the constant capacitance regardless of D_{ctrl} . To further suppress the supply ringing caused by the wirebonding inductance (i.e., L_{bond}), the on-chip decoupling scheme illustrated in Fig. 5(b) includes a small de'Q-ing resistor R_{deQ} , decoupling capacitance C_{decap} near the pads, and local decoupling capacitance $C_{decap,local}$ close to the DTC (stabilizing the local supply and filtering the high-frequency noise). Specifically, R_{deQ} is set in series with L_{bond} to significantly suppress the ringing, given that the DTC does not require any dc biasing currents⁷.

Fig. 6 plots the post-layout simulated INL of an 8-bit DTC and the proposed "half-sine"-shaped fitting curve as $0.35 \times |\sin(k \cdot \pi/255 - 0.1\pi)| - 0.1$ (also see [18], [19]). Accordingly, the INL of the DTC is estimated as 0.35 LSB. Assume that the 8-bit DTC supports a 1st-order DSM-based fractional- N architecture, whereas it must be extended to 9 bits to support a 2nd-order DSM-based architecture. However, the simulated INL of the 9-bit DTC worsens up to ~ 1.5 LSB. Thus, a quantitative analysis is necessary to determine whether the 2nd-order DSM-based architecture could achieve better spur performance.

B. Modeling Frac.- N Spurs by 1st-Order DSM

The transfer function of a K -bit DTC is assumed to be (see Fig. 7(a))

$$\Delta t_d[k] = k \cdot \Delta t_{dte} + T_{inl} \cdot \left| \sin \left(\frac{\pi}{2^K - 1} \cdot k \right) \right| \quad (13)$$

⁷For an analog circuit requiring a dc biasing current, R_{deQ} would be placed in series only with C_{decap} (but not with $C_{decap,local}$) to prevent the IR drop while simultaneously suppressing the supply ringing ($C_{decap,local} = \frac{1}{5} \dots \frac{1}{10} C_{decap}$).

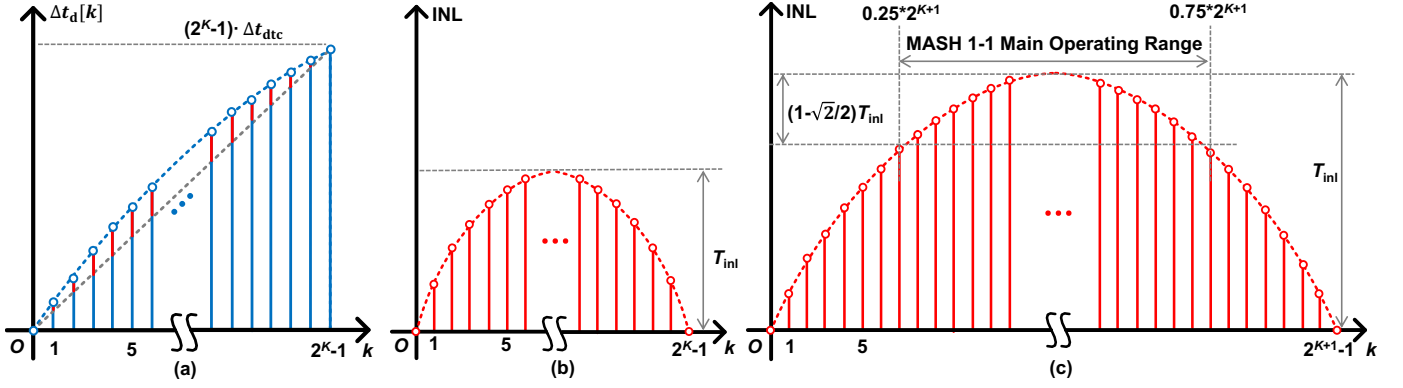


Fig. 7. (a) Transfer function of a K -bit DTC with resolution of Δt_{dtc} , and (b) its INL modeled by a “half-sine” function with a peak-to-peak amplitude of T_{inl} . (c) Maintaining the same Δt_{dtc} , the T_{inl} worsens when extending the DTC from K to $(K + 1)$ bits.

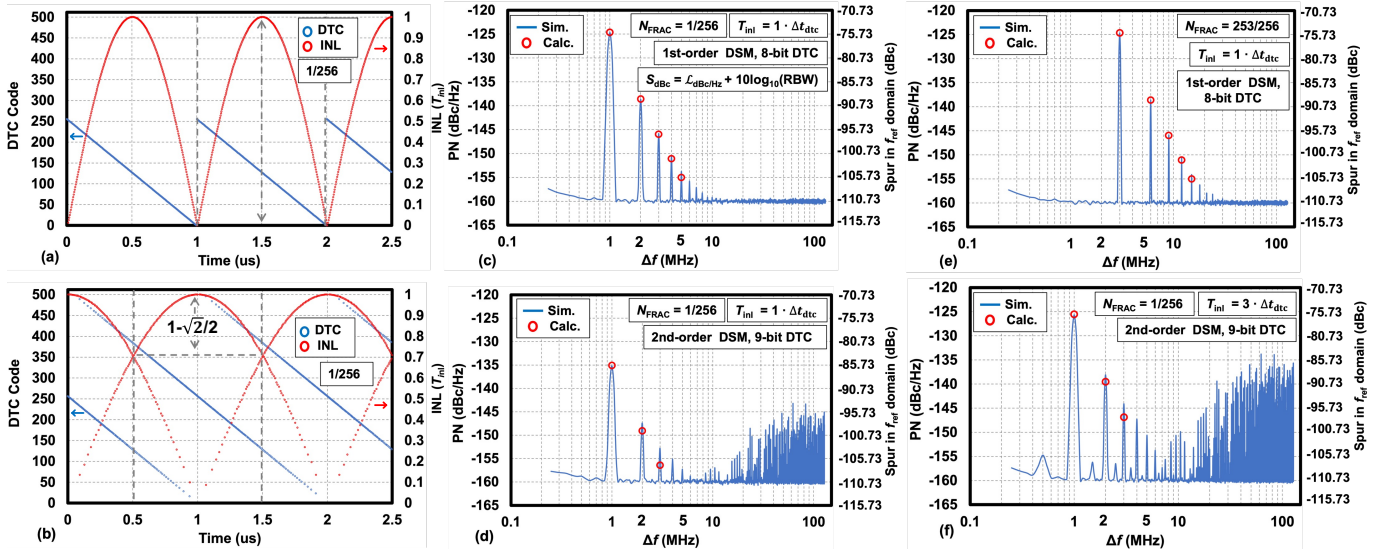


Fig. 8. DTC code change and corresponding INL with (a) 1st-order and (b) 2nd-order DSM-based DTC controllers. Calculated and simulated PN and spur in the DTC output with 1st-order DSM-based DTC controller when $T_{inl} = \Delta t_{dtc}$ with (c) $N_{frac} = 1/256$ and (e) $N_{frac} = 253/256$. Calculated and simulated PN and spur in the DTC output with 2nd-order DSM-based DTC controller when $N_{frac} = 1/256$ with (d) $T_{inl} = \Delta t_{dtc}$ and (f) $T_{inl} = 3\Delta t_{dtc}$. Conditions: $f_{ref} = 256$ MHz, $\Delta t_{dtc} = 500$ fs, $L_{ref} = 160$ dBc/Hz. Frequency bin (f_{bin}) set at 50 kHz; RBW = $1.69 \times f_{bin}$ using Exact-Blackman window.

where T_{inl} (unit: s) is the peak-to-peak amplitude of the INL, and $0 \leq k \leq 2^K-1$. A “half-sine” function is used to model the INL, as discussed in Section IV-A. Assuming $N_{frac} = 1/256$ and $T_0/\Delta t_{dtc} = 2^K = 256$ for simplicity, the DTC control code generated by the 1st-order DSM is illustrated in Fig. 8(a). Consequently, the deterministic jitter of the DTC’s output resulting from its INL, expressed as $\Delta t_d[k] - k\Delta t_{dtc}$, can be approximated as a continuous-time “half-sine” waveform of Fig. 8(a). The corresponding Fourier series is expressed as

$$\begin{aligned} \Delta t_{DJ,1st-dsm}(t) &\approx T_{inl} \cdot |\sin \pi f_{frac} t| \\ &= \frac{2T_{inl}}{\pi} + \sum_{m=1}^M \frac{4T_{inl}}{(4m^2-1)\pi} \cos(m \cdot 2\pi f_{frac} t - \pi). \end{aligned} \quad (14)$$

Therefore, the fractional- N spur observed at offset $m f_{frac}$ in the spectrum of DTC’s output, per (7), is

$$S_{m,dtc} = \left| \pi \cdot \frac{4T_{inl}}{(4m^2-1)\pi} / T_{ref} \right|^2 = \left| \frac{4}{(4m^2-1)} \cdot \frac{T_{inl}}{T_{ref}} \right|^2. \quad (15)$$

Just like the reference PN, $S_{m,dtc}$ undergoes a low-pass filtering due to the PLL. The worst in-band fractional- N spur can be calculated as

$$S_{1st-dsm,half-sine} = N^2 \cdot S_{m,dtc}|_{m=1} = \left| \frac{4}{3} \cdot \frac{T_{inl}}{T_0} \right|^2. \quad (16)$$

C. Modeling Frac- N Spurs With 2nd-Order DSM & INL Shaping by MASH 1-1 Modulator

The DTC controller based on a 2nd-order DSM requires the DTC’s coverage of up to $2T_0$, resulting in an inadvertent extension of the DTC from K to $(K + 1)$ binary bits with the same Δt_{dtc} . Such an extension statistically worsens the INL, as illustrated in Fig. 7(c). Unlike in the 1st-order DSM, the DTC delay controlled by the 2nd-order DSM follows the basic $[1 - (nN_{frac} - \lfloor nN_{frac} \rfloor)] \cdot T_0$ but with stochastically scaled $+T_0$ (see (3)), as shown in Fig. 8(b) (also see Fig. 3 in [33] for further details).

For one fractional- N period, $1/f_{frac}$ [e.g., from $0.5 \mu s$ to $1.5 \mu s$ in Fig. 8(b)], around 80% of the DTC control codes

concentrate within the middle range (in the example shown: codes decreasing from $0.75 \times 2^{K+1}$ down to $0.25 \times 2^{K+1}$, with $K = 8$). It results in the delay from $1.5T_0$ to $0.5T_0$, which differs from $T_0 \rightarrow 0$ with the 1st-order DSM. Consequently, the shaped INL can be effectively modeled by its repeated middle range under the control of MASH 1-1, as illustrated in Fig. 9.

1) *Half-Sine INL Shaped as Half-Sine*: Interestingly, the middle range of the half-sine INL can be modeled by another half-sine but with the peak-to-peak amplitude reduced to $(1 - \sqrt{2}/2)T_{\text{inl}}$, shown in Fig. 9(a)⁸. Thus, the corresponding deterministic jitter in Fig. 8(b) can be estimated as (blue line in Fig. 9(a))

$$\Delta t_{\text{DJ,MASH1-1}}(t) \approx \frac{\sqrt{2}}{2}T_{\text{inl}} + (1 - \frac{\sqrt{2}}{2})T_{\text{inl}} \cdot \left| \sin(\pi f_{\text{frac}} t + \frac{\pi}{2}) \right|. \quad (17)$$

Its in-band spur is derived analogously to (16), resulting in

$$S_{\text{MASH1-1,half-sine}} \approx \left| \frac{4}{3} \cdot \frac{(1 - \sqrt{2}/2)T_{\text{inl}}}{T_0} \right|^2. \quad (18)$$

By comparing (16) and (18), we conclude that, given the same Δt_{dtc} resolution step, if the middle range INL of the $(K+1)$ -bit DTC (i.e., DTC delays from $1.5T_0$ to $0.5T_0$) is better than the INL of the K -bit DTC (i.e., DTC delays from T_0 to 0), then the 2nd-order DSM-based fractional- N architecture with the $(K+1)$ -bit DTC will *improve* the spur performance. In this sense, adding a code offset in the 1st-order DSM-based controller to utilize the middle range of the $(K+1)$ -bit DTC (with the $2T_0$ coverage) can achieve the spur performance similar to that of a MASH 1-1-based architecture; this demystifies the MASH 1-1 approach.

2) *Sine INL Shaped as Sawtooth*: The proposed spur analysis technique with the MASH 1-1 modulator can also be applied to other INL shapes. Recently, a pseudo-differential DTC has shown a low sine-shaped INL by differentiating two large half-sine INLs [58]. As shown in Fig. 9(b), for such a sine INL with peak-to-peak amplitude T_{inl} , its repeated middle range can be modeled by a sawtooth shape with a peak-to-peak amplitude $12/\pi^2 \cdot T_{\text{inl}}$ using the least mean square curve-fitting principle. Based on (7), the in-band spur can be derived as

$$S_{\text{MASH1-1,sine}} = \left| \pi \cdot \frac{12/\pi^2 \cdot T_{\text{inl}}}{\pi} / T_0 \right|^2 = \left| \frac{12}{\pi^2} \cdot \frac{T_{\text{inl}}}{T_0} \right|^2 \quad (19)$$

where $(12/\pi^2 \cdot T_{\text{inl}})/\pi$ is the magnitude of the fundamental frequency component of the sawtooth-like INL.

It is important to note that the aforementioned quantitative analysis of spurs can be extended to voltage-domain fractional- N synthesizers [11], [40]–[43], [46] by translating the nonlinearity of the DAC (or ADC) from the voltage domain to time domain through either computation or simulation.

⁸It can be verified that $y = \sin(x + \pi/4) - \sqrt{2}/2$ can be approximately by $(1 - \sqrt{2}/2)\sin 2x$ when $0 < x < \pi/2$.

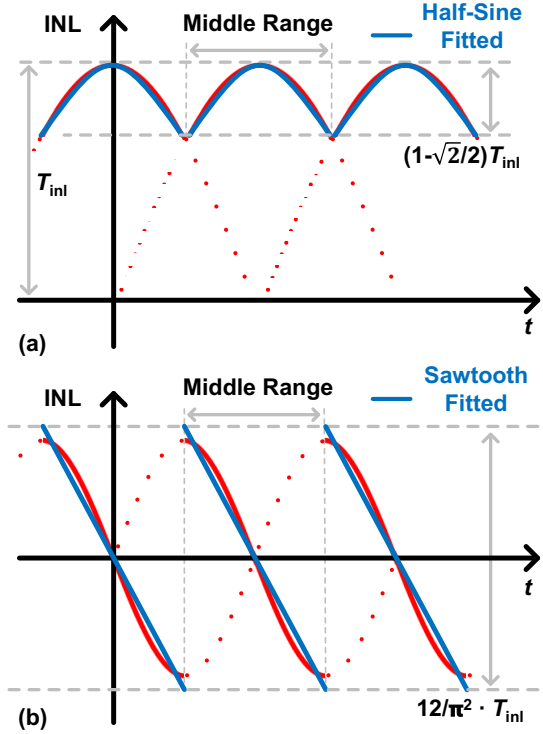


Fig. 9. Diagram of INL-shaping due to the MASH 1-1 modulator: (a) half-sine INL shaped as half-sine, (b) sine INL shaped as sawtooth.

D. Numerical Verification Using Time-Domain Behavioral Simulation

To verify the proposed fractional- N spur formulas for DTC-based frequency synthesizers, we compare them with simulation results from a behavioral model implemented in Cadence Spectre AMS Designer. These verifications were conducted by directly checking both the random and deterministic jitter of the DTC, which were recorded and post-processed in MATLAB to report PN [8], [59]–[61] and spurious content. The spurs in the PN plots are calculated as follows:

$$S_{\text{dBc}}(\Delta\omega) = \mathcal{L}_{\text{dBc/Hz}}(\Delta\omega) + 10 \cdot \log_{10} \text{RBW} \quad (20)$$

where $\mathcal{L}_{\text{dBc/Hz}}(\Delta\omega)$ represents the spur level in dBc/Hz at $\Delta\omega$ before normalized by the resolution bandwidth (RBW). The RBW here depends on the chosen frequency bin (i.e., f_{bin} , unit frequency in power spectrum) and window function⁹. For an accurate spur simulation, the f_{bin} 's harmonics must coincide with the spur locations.

The simulated and calculated spurs with the 1st-order DSM controller at different N_{frac} values are shown in Fig. 8(c) and (e), demonstrating the efficacy of (12) and (15). Fig. 8(d) and (f) show the calculated and simulated spurs with the 2nd DSM for different T_{inl} . Based on (15) and (17), we can accurately estimate the 2nd-order DSM-based fractional- N in-band spurs but may slightly underestimate the spurs at higher harmonics due to the omission of some codes in Fig. 8(b). Fortunately, the fractional- N spurs at higher harmonics do not significantly

⁹For example, using the Exact-Blackman window (with a noise power bandwidth factor of 1.69 [62]) for the timestamps, we get $\text{RBW} = f_{\text{bin}} \times 1.69$.

affect the total rms jitter since they are further low-pass filtered by the PLL.

E. Numerical Verification with Measurements from Literature

To further validate the efficacy of the derived formulas, we compare the calculated results with measurements from the recent literature that involve different INL shapes and $\Delta\Sigma$ quantization cancellation techniques [19], [21], [46], [58], [63], as summarized in Table II.

The measured in-band fractional- N spurs in [21], [63] can be verified against the theory using (18), while those in [58] can be verified using (19). Specifically, with the introduction of the DTC's range-reduction technique in [19], the DTC codes are concentrated in the 400–600 range, exhibiting a half-sine INL estimated at 0.1 LSB, and whose spurs can be calculated using (16).

The proposed analysis technique can also be applied to the voltage-domain of $\Delta\Sigma$ cancellation [46], where the DAC INL shape transitions from a sine wave to sawtooth under the control of a MASH 1-1 modulator. After translating from the voltage- to time-domain with $T_{\text{inl}} = 2011 \text{ fs}^{10}$, the spurs are calculated using (19).

The difference between the calculated and experimental results is well within 1.5 dB, which demonstrates remarkable accuracy of the proposed spur analysis technique.

V. QUANTITATIVE ANALYSIS OF REFERENCE SPURS

A. System Requirements for Reference Spurs

Reference spurs, which occur at the reference rate, can substantially degrade the jitter profile according to the following formula for the total rms jitter:

$$J_{\text{rms,total}} = \frac{\sqrt{2 \cdot (10^{\text{IPN}_{\text{dBc}}/10} + \Sigma 10^{S_{\text{dBc,frac}}/10} + 10^{S_{\text{dBc,ref}}/10})}}{2\pi f_0} \quad (21)$$

where $S_{\text{dBc,frac}}$ and $S_{\text{dBc,ref}}$ represent the relative power of fractional- N and reference spurs, respectively. Thus, when targeting low-jitter requirements, the reference spur should also satisfy (1).

Another adverse effect of reference spurs is their potential mix with adjacent channels¹¹, leading to an adjacent channel interference in a receiver, as illustrated in Fig. 10. To mitigate such an impact on the signal-to-noise ratio (SNR), the reference spur should meet

$$S_{\text{dBc,adj}} + S_{\text{dBc,ref}} \leq -\text{SNR} - 10\text{dB} \quad (22)$$

where $S_{\text{dBc,adj}}$ is the relative power level of an adjacent channel. Given $\text{SNR} = 25 \text{ dB}$ (i.e., induced $\text{EVM}_{\text{SNR}} = 5.5\%$) for a 64-QAM 5G modulation signal with $S_{\text{dBc,adj}} = 28 \text{ dBc}$ [1], the required $S_{\text{dBc,ref}} = -63 \text{ dBc}$.

¹⁰ $T_{\text{inl}} = 112 \text{ mV} \times \frac{431 \text{ fs}}{24 \text{ mV}} = 2011 \text{ fs}$, where 112 mV is the peak-to-peak INL of DAC, and $\frac{431 \text{ fs}}{24 \text{ mV}}$ is the extracted gain from voltage-domain to time-domain in [46].

¹¹For example, in an integer- N PLL, the reference frequency is chosen to match the channel spacing of the wireless communication system.

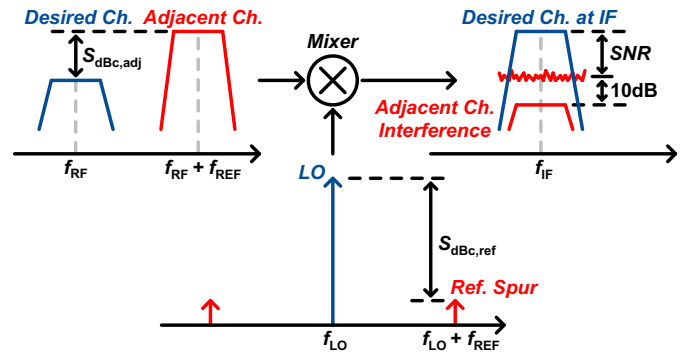


Fig. 10. Reference spur causes adjacent channel interference and its requirement to ensure no significant impact on SNR.

B. Reference Spurs in CP-PLLs

Reference spurs in CP-PLLs are a type of FM-induced spurs resulting from ripples in the tuning voltage (V_{tune}) of the oscillator. These ripples are intrinsically caused by imperfections in the charge-pump, such as mismatch between the up and down currents, which dominate over the required “anti-backlash pulses¹²” [52]. By leveraging the Fourier series of V_{tune} (which can be modeled as a sawtooth waveform) and the VCO gain of K_{VCO} (unit: Hz/V), the reference spurs in CP-PLLs can be calculated directly using (11) as

$$S_{m,\text{ref-PLL}} = \left| \frac{1}{2} \frac{K_{\text{VCO}} \cdot V_{\text{saw}}/m\pi}{mf_{\text{ref}}} \right|^2 = \left| \frac{K_{\text{VCO}} \cdot V_{\text{saw}}}{2\pi m^2 f_{\text{ref}}} \right|^2 \quad (23)$$

where V_{saw} is the peak-to-peak value of sawtooth-like V_{tune} and $V_{\text{saw}}/m\pi$ is the magnitude of m th harmonic of V_{tune} . The higher harmonic spurs for $m > 1$ are attenuated by a factor of $1/m^4$, exhibiting a -40 dB/dec slope. When $m = 1$ or $m = 2$, the expression reduces to the corresponding equations in [52].

It is worth noting that there will be no additional loop filtering for such reference spurs since they arise from the loop itself, correcting the imperfections of the charge pump. As for the other FM-induced spurs caused by supply ripples directly affecting the oscillator, such spurs will be filtered out by the synthesizer's high-pass transfer function, similar to the filtering of the oscillator's intrinsic PN.

C. Reference Spurs in Injection Locking

The DTC-based fractional- N spurs in injection-locked synthesizers [39] are similar to those in PLLs. In this subsection, we focus on the reference spur due to injection locking. Such a spur is a type of PM-induced spurious tone, inherently caused by the difference between the oscillator's free-running period (T_{osc}) and the forced locked period (T_0). Assuming $t_{\text{out}}[k]$ is the output timestamp of the IL system, the instantaneous deterministic jitter can be derived as $\Delta t_{\text{DJ,IL}}[k] = t_{\text{out}}[k] - k \cdot T_0$. This jitter is maximized at the $(N - 1)$ th cycle before the full correction at the N th cycle, as illustrated in Fig. 11(a). For simplicity and without any loss of generality, $\Delta t_{\text{DJ,IL}}[k]$ can be further approximated as a continuous-time sawtooth

¹²Anti-backlash pulses refer to extremely narrow pulses between the up and down currents of a charge-pump, used to eliminate the deadzone when the PLL is locked.

TABLE II

COMPARISON OF CALCULATED AND MEASURED FRAC.- N SPURS WITH DIFFERENT INL SHAPES AND $\Delta\Sigma$ QUANTIZATION CANCELLATION TECHNIQUES.

	Wu, JSSC'21 [19]	Dartizio, JSSC'22 [63]	Xu, ISSCC'24 [58]	Rossoni, ISSCC'24 [21]	Kim, JSSC'24 [46]
Technology	14nm FinFET	28nm CMOS	65nm CMOS	28nm CMOS	65nm CMOS
Architecture	Analog sampling PLL	Digital sampling BBPLL	Digital sampling BBPLL	Digital sampling BBPLL	Digital sub-sampling PLL
Frequency (GHz) ($1/T_0$)	5.376	13	7	8.75	14
Order of DSM	MASH 1-1 (w/ DTC range reduction)	MASH 1-1	MASH 1-1	MASH 1-1	MASH 1-1
$\Delta\Sigma$ Quant. Cancellation	Time-domain	Time-domain	Time-domain	Time-domain	Voltage-domain
Peak-to-peak INL (fs) (T_{inl})	35* ^a	500*	580**	1000**	2011*
Shape of INL	Half-Sine	Half-Sine	Sine	Half-Sine	Sine
Meas. Frac. Spur (dBc)	-73	-51.1	-45.4 (Cascaded Divider Technique OFF)	-50.9 (ACZ OFF)	-29 (SCF-DPD OFF)
Calc. Frac. Spur (dBc)	-72	-51.9	-46.1	-49.3	-29.3

*Reported or estimated by fitting the simulated results with half-sine or sine shape

**Reported or estimated by fitting the measured results with half-sine or sine shape

^aEstimated by 0.1 LSB

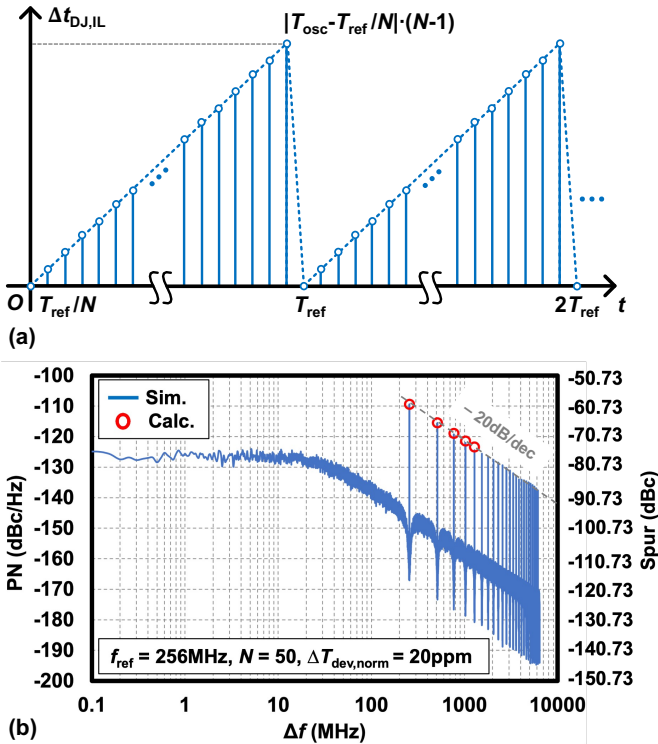


Fig. 11. (a) Deterministic jitter introducing reference spur. (b) Calculated and simulated PN and reference spur. Conditions: $\beta=0.5$, $f_{ref} = 256\text{ MHz}$, $N = 50$, $\mathcal{L}_{ref} = 10^{-160/10}\text{ rad}^2/\text{Hz}$, and $\mathcal{L}_{osc} @ 10\text{ MHz} = 10^{-140/10}\text{ rad}^2/\text{Hz}$. Frequency bin (f_{bin}) set at 50 kHz ; RBW = $1.69 \times f_{bin}$ using the Exact-Blackman window.

waveform, shown in Fig. 11(a). Thus, the reference spur at the $m f_{ref}$ offset is calculated by (7)

$$S_{m,ref-IL} = \left| \pi \cdot \frac{T_{saw}}{m\pi} / T_0 \right|^2 = \left| \frac{T_{saw}/T_0}{m} \right|^2 \quad (24)$$

where $T_{saw} (= |(T_{osc} - T_0) \cdot (N - 1)|)$ is the peak-to-peak magnitude of the sawtooth waveform, and $T_{saw}/m\pi$ is the magnitude of m th harmonic of sawtooth-like $\Delta t_{DJ,IL}$. The higher harmonic spurs for $m > 1$ are attenuated by a factor of $1/m^2$, exhibiting a -20 dB/dec slope, differing from those in CP-PLLs. Similarly to the reference spurs in CP-PLLs, the reference spurs discussed here originate from the IL's intrinsic correction mechanism; thus, there is no additional loop filtering.

Assuming the normalized oscillator's period deviation $\Delta T_{dev,norm} = |(T_{osc} - T_0)/T_0| = 20\text{ ppm}$, we now validate the reference spur formula (24). For instance, considering a 256 MHz reference with $\mathcal{L}_{ref} = 10^{-160/10}\text{ rad}^2/\text{Hz}$ performing injection locking of a 12.8 GHz oscillator (i.e., $N = 50$) with $\mathcal{L}_{osc} @ 10\text{ MHz} = 10^{-140/10}\text{ rad}^2/\text{Hz}$ in a free-running mode will yield the PN and spur plot depicted in Fig. 11(b). The analytical results obtained from (24) [i.e., red circles] closely match with the simulation results.

VI. CONCLUSION

This paper presents a unified quantitative time-domain spur analysis method that inherently distinguishes between spurs induced by instantaneous phase modulation and those caused by frequency modulation. Using the proposed INL-shaping modeling technique, we compare the INL-induced fractional- N spur with 1st- and 2nd-order DSMs and numerically verify the proposed analytical results through time-domain behavioral simulations and measurements from the literature. We further provide a comprehensive guide for understanding and choosing the architectures for $\Delta\Sigma$ quantization cancellation techniques. Reference spurs in both PLLs and injection-locked synthesizers are incorporated into the theoretical framework. The introduced spur analysis technique can extend to other fractional- N synthesizers of low-spur performance.

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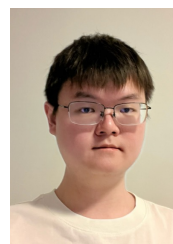
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